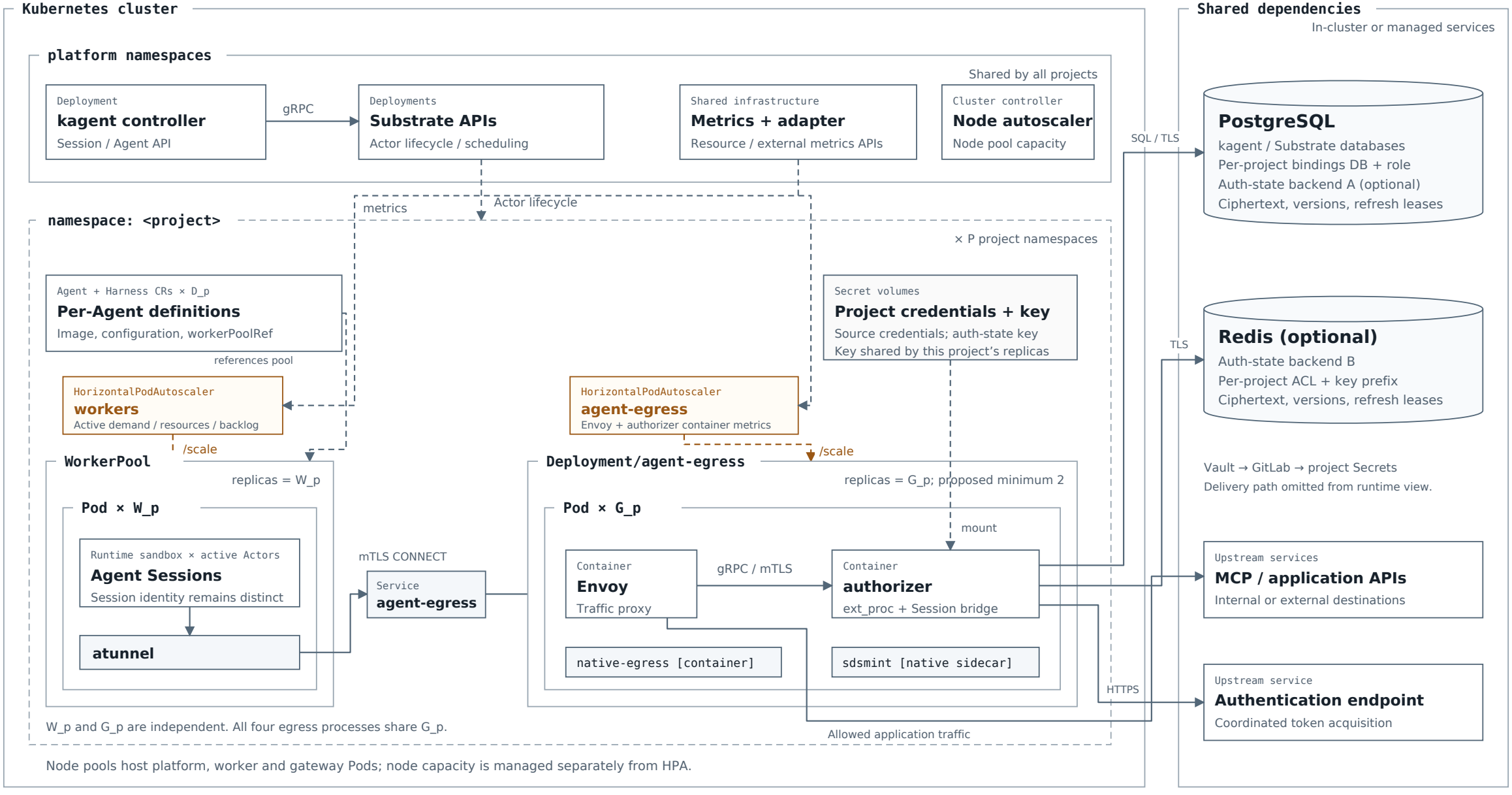
Kubernetes cluster
platform namespaces
Shared by all projects
Deployment
kagent controller
Session / Agent API
gRPC
Deployments
Substrate APIs
Actor lifecycle / scheduling
Shared infrastructure
Metrics + adapter
Resource / external metrics APIs
Cluster controller
Node autoscaler
Node pool capacity
metrics
Actor lifecycle
namespace: <project>
× P project namespaces
Agent + Harness CRs × D_p
Per-Agent definitions
Image, configuration, workerPoolRef
references pool
Secret volumes
Project credentials + key
Source credentials; auth-state key
Key shared by this project's replicas
HorizontalPodAutoscaler
workers
Active demand / resources / backlog
/scale
HorizontalPodAutoscaler
agent-egress
Envoy + authorizer container metrics
/scale
WorkerPool
replicas = W_p
Pod × W_p
Runtime sandbox × active Actors
Agent Sessions
Session identity remains distinct
atunnel
mTLS CONNECT
Service
agent-egress
Deployment/agent-egress
replicas = G_p; proposed minimum 2
Pod × G_p
mount
Container
Envoy
Traffic proxy
gRPC / mTLS
Container
authorizer
ext_proc + Session bridge
native-egress [container]
sdsmint [native sidecar]
W_p and G_p are independent. All four egress processes share G_p.
Allowed application traffic
Node pools host platform, worker and gateway Pods; node capacity is managed separately from HPA.
Shared dependencies
In-cluster or managed services
SQL / TLS
PostgreSQL
kagent / Substrate databases
Per-project bindings DB + role
Auth-state backend A (optional)
Ciphertext, versions, refresh leases
TLS
Redis (optional)
Auth-state backend B
Per-project ACL + key prefix
Ciphertext, versions, refresh leases
Vault → GitLab → project Secrets
Delivery path omitted from runtime view.
Upstream services
MCP / application APIs
Internal or external destinations
HTTPS
Upstream service
Authentication endpoint
Coordinated token acquisition